

PAPER_950: BCS Critical Temperature in SCm Vacuum

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** bcs_superconductivity_uqff.py (BCSCriticalTemperature) **Calculator:** BCSCriticalTemperatureCalc (CP4 #534) **CVW:** v2.0.0 compliant

Abstract

We compute the BCS critical temperature T_c in the SCm vacuum phonon framework. The relation $T_c = (1.13 \cdot \hbar\omega_{\text{SCm}}/k_B) \cdot \exp(-1/N(0)V_{\text{SCm}})$ replaces the conventional Debye cutoff with $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz, yielding critical temperatures governed by the SCm phonon attraction strength V_{SCm} and Fermi-level density of states $N(0)$.

1. Critical Temperature

$$T_c = \frac{1.13\hbar\omega_{\text{SCm}}}{k_B} \exp\left(-\frac{1}{N(0)V_{\text{SCm}}}\right)$$

2. BCS Gap Relation

$$\Delta(0) = 1.764 k_B T_c$$

3. Source Data

- **File:** bcs_superconductivity_uqff.py
 - **Session:** 214
 - **CP4 Class:** BCSCriticalTemperatureCalc (#534)
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References

1. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)
2. Bardeen, J., Cooper, L.N. & Schrieffer, J.R. (1957) - Phys. Rev. 108, 1175